At a matched budget, shape is free: only size buys anything

N=48, degree 5, 4 colours, leak-free | two attention-only rungs 2.6% apart in parameters | J 51280737, 6 cells, 3 seeds each

THE CONTRAST, AND IT IS AS CLEAN AS THIS GRID CAN MAKE IT

Shallow-wide (L=12 d224, 2,571,130) lands at rho_50 0.579; deep-narrow (L=16 d192, 2,504,890) at 0.580. The difference is 0.001 = 0.00 s* against a frozen bar of 0.1484. Both cross on 3 of 3 seeds and clear finding 44's 1.64 plateau on 3 of 3. Two shapes, one price, one answer: over this grid model quality is ONE-DIMENSIONAL, and the depth and width ladders were reading the same axis at different prices.

THE STANDING MINIMUM MOVES, AND THE CLAIM IS 'SMALLER', NOT 'BETTER'

The registered rule (3 of 3 crossing AND median at or below 0.74) fires for BOTH rungs, so the tie-break named at scoring time -- smaller parameter count -- gives J2. The minimum goes from attention-only L=12 d256 at 3,331,642 / 0.605 to attention-only L=16 d192 at 2,504,890 / 0.580: 1.33x smaller, 1.96x below the hub, rho_50 moving -0.025 = 0.17 s* -- below the bar. The two J rungs differ by less than the job's own 5% tolerance, so the real statement is that the minimum is ~2,500,000 AT EITHER SHAPE.

THE PREDICTION THAT MISSED, REPORTED FIRST BECAUSE IT MISSED

J1 was registered to land in [0.665, 0.69], from three interpolations of its neighbours (linear in d 0.68, linear in trainable 0.685, linear in 1/d 0.669). It came in at 0.579 -- MISSED LOW, all three falsified in the same direction. d224 beats any straight line through d192 and d256, so the width curve is CONVEX there: the gain is bought in the first 32 units of width, not the last. Nothing on this axis should be interpolated again without a fourth point.

WHAT IS STILL NOT CLAIMED

The best solver on this eval at any size is still the COMPLETE model at L=12, rho_50 0.385 at 9,638,458 trainable; 'minimal' has never meant 'best'. Every number here is N=48 -- whether any of it survives a second problem size is stage F, running now as 51656823.

rho_50 = realized hint density at which solve rate crosses 0.5 on a 14-point x 400-instance grid, LOWER IS BETTER, reported CENSORED when no bin crosses. mean_solve = unweighted mean over the 14 bins, which never censors. s* = 0.1484 / 0.2918 was frozen before any of these cells ran, and the free line 0.74 is the hub plus one s*. Commit (uncommitted).

What the model actually sees

Every token is one categorical id into a 58-symbol vocabulary (N=48, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING
	BEGIN_ROUND SOLVED UNRESOLVED
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	COLOR_0 COLOR_1 COLOR_2 COLOR_3
	the q colours (token name COLOR_n, prose 'colour')
10-57	N0 N1 ... N47
	one id per node, no structure in the id

GRAMMAR

BEGIN_HINTS	(NODE COLOR)*
BEGIN_GRAPH	(NODE (NODE){degree})*
BEGIN_REASONING	(BEGIN_ROUND (NODE COLOR)+)*

SOLVED

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image: node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 401 TOKENS

HINTS	BEGIN_HINTS N0 COLOR_0 N1 COLOR_2 N2 COLOR_3 N10 COLOR_1 N15 COLOR_1 N17 COLOR_3 ... (37 tokens)
GRAPH	BEGIN_GRAPH N0 N16 N8 N38 N34 N17 N1 N13 N27 N34 N28 N10 ... (289 tokens)
REASONING	BEGIN_REASONING BEGIN_ROUND N9 COLOR_3 N14 COLOR_2 N16 COLOR_2 N32 COLOR_0 N34 COLOR_1 N44 ... (75 tokens)

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 13 rounds, 30 deductions,
60 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000018, FALL-colouring ensemble (every node sees all q-1 other colours, so no node is permanently undeducible).

One real reasoning trace

18 hints of 48 nodes, then 13 rounds of simultaneous deduction. Hints: N0=COLOR_0 N1=COLOR_2 N2=COLOR_3 N10=COLOR_1 N15=COLOR_1 N17=COLOR_3 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND      N9 COLOR_3 N14 COLOR_2 N16 COLOR_2 N32 COLOR_0 ...
  6 deduced
BEGIN_ROUND      N42 COLOR_3 N45 COLOR_1
  2 deduced
BEGIN_ROUND      N13 COLOR_0 N20 COLOR_0 N37 COLOR_2
  3 deduced
BEGIN_ROUND      N21 COLOR_2 N24 COLOR_3 N31 COLOR_2
  3 deduced
... 9 further rounds

SOLVED
```

WHY ONE DEDUCTION FIRES

node N9, neighbours N25 N10 N29 N31 N18

their colours known before this round:

```
[0, 1, 2]  (3 = q-1 of the 4)

(3 known neighbours, 3 DISTINCT colours)

still unknown: N25 N31
```

only colour left: 3

```
so the trace emits  N9 COLOR_3
```

The model is never shown this reasoning -- only the node/colour pair.

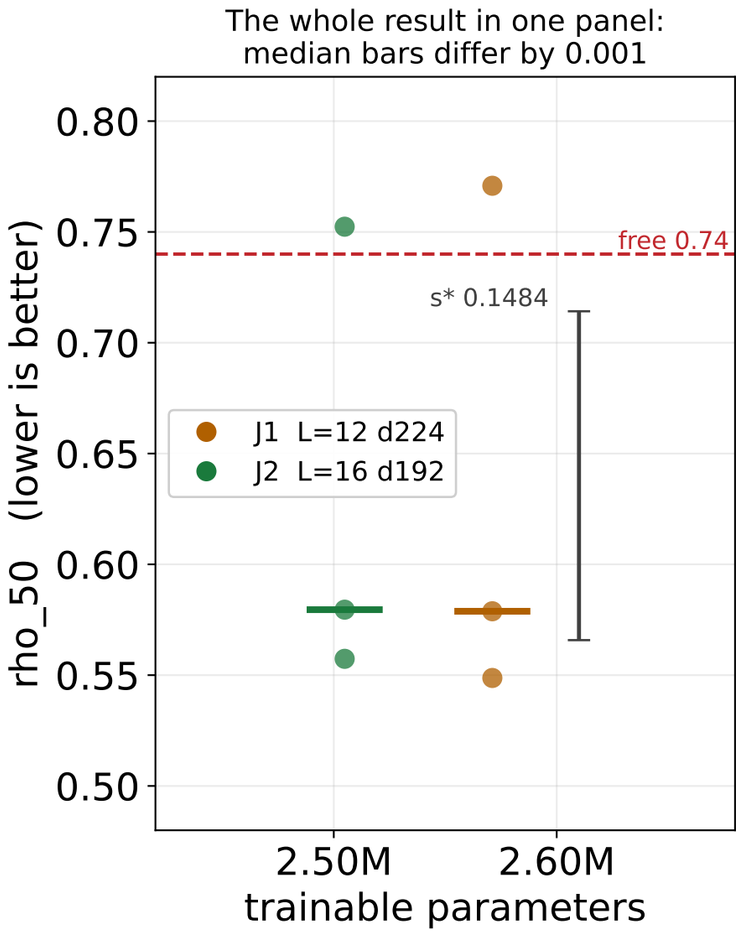
A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

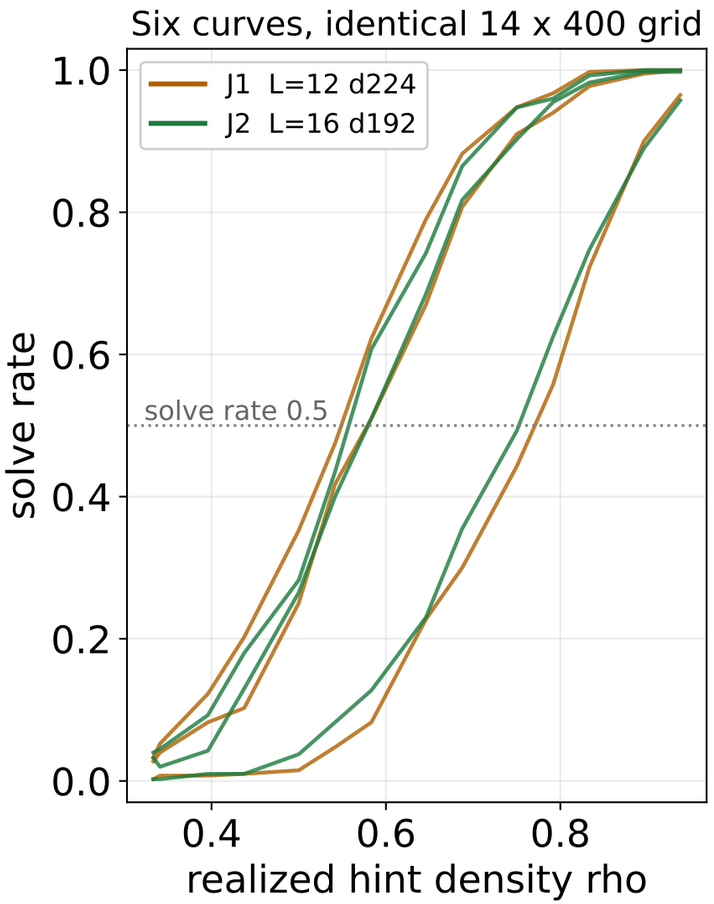
N=48, degree 5, 4 colours, --shuffle-adj --fall: the task every cell in this deck is trained and scored on. Stage J changes nothing about it -- only num_layers and d_model move, and they move against each other so the two budgets land 2.6% apart, inside the 5% tolerance the contrast was registered under.

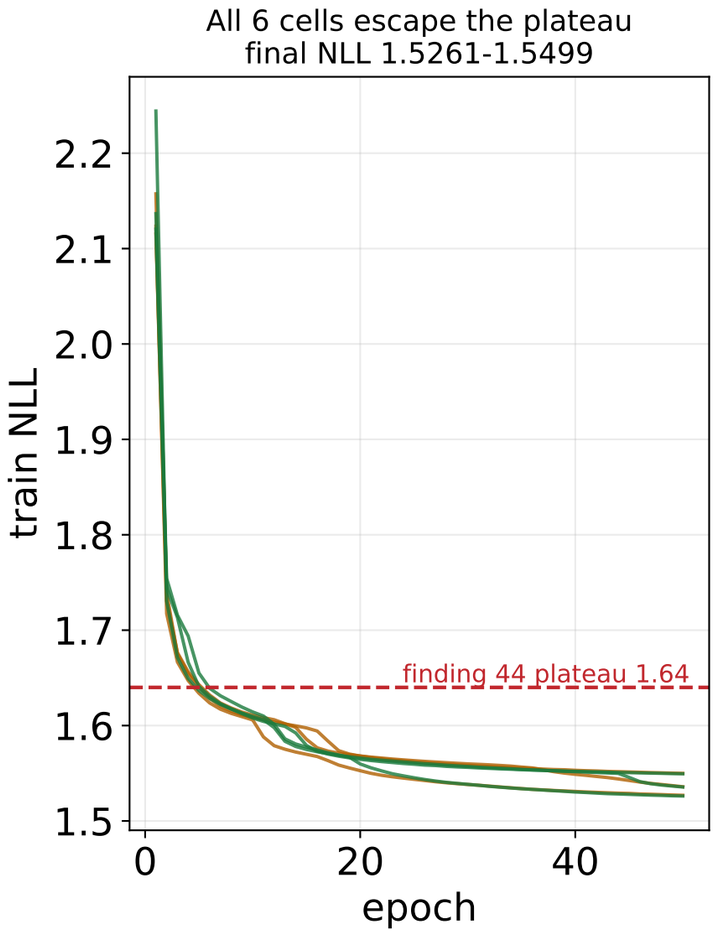
T=13 rounds, the 90th percentile of 40 draws from this config (range 1-16) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

Two shapes at one price, and they are indistinguishable

Left: every seed, not just the median. Middle: the curves rho_50 is read off. Right: training, where the two arms also coincide



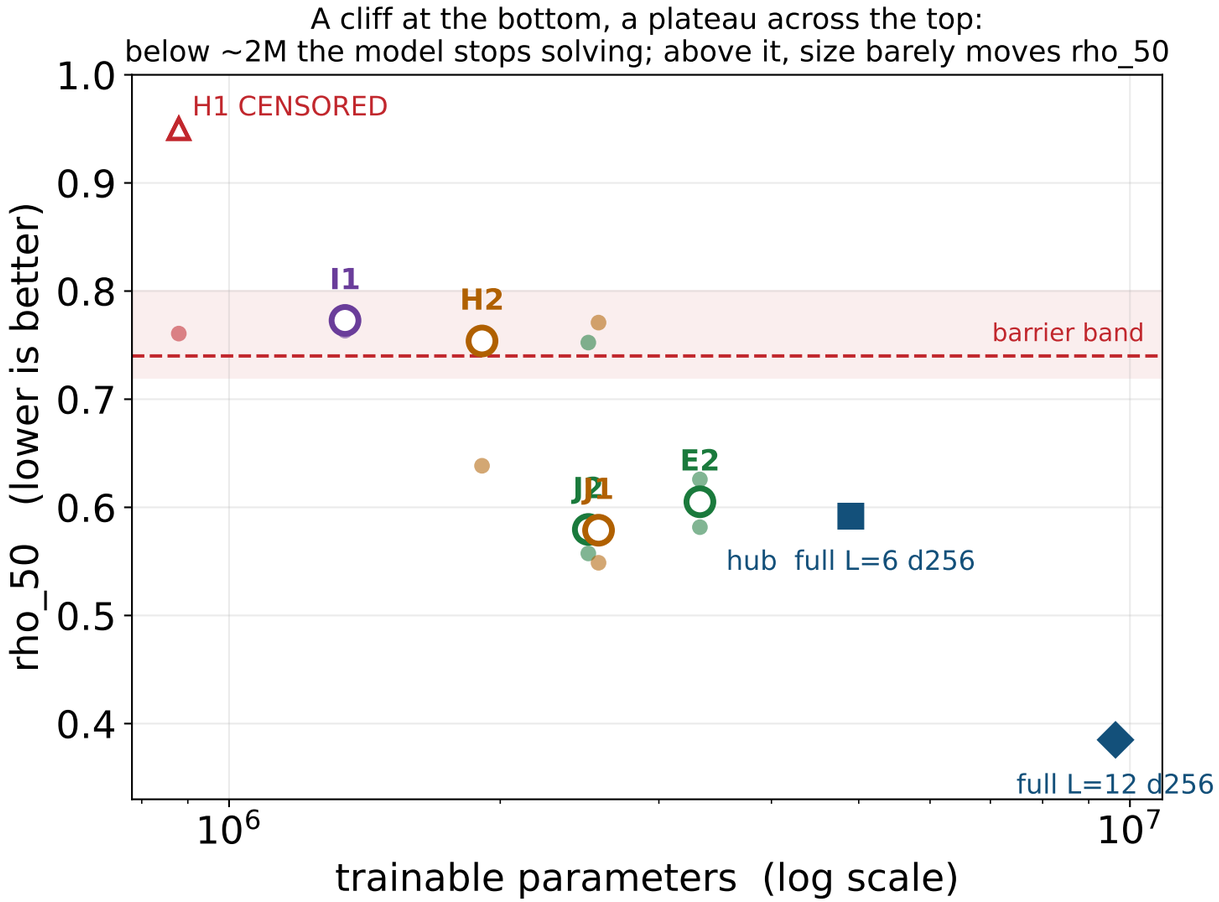




Open triangles would mark censored seeds; there are none in stage J. The median bar is the 3-seed median rho_50, the quantity every verdict is read on. Both arms were trained on the same 240k-instance pool for 50 epochs at constant lr 1e-3, batch 32, no warmup -- the only things that differ between them are num_layers and d_model. Commit (uncommitted).

The size-quality frontier, and where stage J put the knee

Every attention-only rung this programme has run at N=48, against the two complete-model anchors



rung	trainable	rho_50	cross	plateau esc
no_mlp L=12 d128	879,418	CENSORED	1/3	1/3
no_mlp L=12 d160	1,345,018	0.773	3/3	3/3
no_mlp L=12 d192	1,908,922	0.754	3/3	3/3
no_mlp L=16 d192	2,504,890	0.580	3/3	3/3
no_mlp L=12 d224	2,571,130	0.579	3/3	3/3
no_mlp L=12 d256	3,331,642	0.605	3/3	3/3
full L=6 d256 (hub)	4,899,898	0.592	3/3	-
full L=12 d256	9,638,458	0.385	3/3	-

WHAT THE SHAPE OF THIS CURVE SAYS. From 3,331,642 down to 2,504,890 -- a 1.33x cut -- rho_50 does not move outside the bar. Below that the curve turns over hard: d192 at L=12 is already in the barrier band, d160 is deeper into it, and d128 does not train on 2 of 3 seeds. The useful reading is not 'smaller is free' but 'there is a floor near 2M parameters on this task, and everything above it is flat'. Stage J's two rungs sit exactly on the last flat step before the drop, which is why the matched contrast is informative: if shape mattered anywhere, it would matter at the edge.

Open triangles are censored rungs, drawn at the grid edge 0.95 because that is all the eval knows about them. All cells attention-only unless marked 'full'; all at 240k instances x 50 epochs, batch 32, constant lr 1e-3, no warmup, --shuffle-adj --fall. Commit (uncommitted).

Every stage J prediction, scored against what was registered

stagej_score.py was committed (88cb7cb) while 51280737 was in its first wave -- no verdict was chosen after seeing a number

J0 HELD -- shape is not a free parameter (registered: |J1-J2| <= s* = 0.1484)

Observed 0.001 = 0.00 s*. The budgets came back 2,571,130 vs 2,504,890, 2.6% apart against the 5% tolerance that would have VOIDED this contrast, so it is a controlled comparison and not a size verdict wearing a hat. The falsifier named a winner in advance in either direction; neither fired.

J1 MISSED LOW (registered: median in [0.665, 0.69] on 3/3 cross, 3/3 escape)

Observed 0.579 on 3/3 crossing and 3/3 escapes -- the counts held, the level did not. All three registered interpolations (linear in d 0.68, linear in trainable 0.685, linear in 1/d 0.669) are falsified in the same direction, which is the informative part: the width curve is convex on [192, 256], so interpolating d_model between measured rungs systematically UNDERESTIMATES the in-between model. No fourth point, no more interpolation.

J2 HELD -- depth past L=12 still pays (registered: below H2's 0.754 on >= 2/3 escapes)

Observed 0.580 on 3/3 escapes, -0.174 = 1.18 s* against H2 at the SAME width -- lower and separable. The depth evidence this rested on stopped at L=12 (L=6 0.763, L=8 0.749, L=12 0.605), so L=16 was an extrapolation of the ladder and it came back on the line. This is the first leak-free point above L=12 at N=48.

J3 FIRED | J4 HELD -- 22/22 held-out rungs on the curve, worst miss 0.0222 vs bar 0.0423

J3's rule (3/3 crossing AND median at or below 0.74) fired for both rungs and the minimum moved to J2 on the pre-named tie-break. J4 refits the depth ladder's accuracy-to-solve curve over all nine scan dirs and asks whether J's rungs sit on it: 22 of 22 well-determined held-out rungs do, over 47 cells, 21 of them discriminating. That is UP from 21/21 over 46 before stage J. Token accuracy remains a sufficient statistic over this grid -- including across a shape change it was never fit on.

The LOO bar 0.0423 is the leave-one-out RMS of the depth ladder's own fit and was frozen with it. 'Discriminating' means the curve predicts solve > 0.10 for that cell, so landing on it is a real constraint rather than two numbers agreeing near zero. Commit (uncommitted).

Where this leaves the programme, and what 51656823 is testing now

stage J closes the N=48 ladder | stage F (51656823) asks whether any of it survives a second problem size

THE N=48 LADDER IS CLOSED, AND THE CLOSING ARGUMENT IS J0

There is no further architecture cut this grid can resolve. Width below d192 does not train (stage H), the d160-d192 stretch buys nothing a seed could not (stage I), and stage J has now shown that trading depth against width at a fixed budget changes nothing measurable. What is left to vary is size, and the frontier is flat from ~2.5M up to the hub's 4,899,898 and falls off a cliff below. The standing minimum is attention-only L=16 d192, 2,504,890 trainable at rho_50 0.580.

STAGE F (51656823): THE ONE QUESTION A SINGLE PROBLEM SIZE CANNOT ANSWER

Every cut above was measured at N=48. F runs the surviving architecture at N=96 against a PAIRED complete-model control -- 4 cells, 36.5 GPU-h, cheap only because the control already exists: 8 N=96 cells trained leak-free at the same 240k x 50 budget on the identical 14 x 400 grid (full L=12 d256 -> 0.621 over two seeds). F buys the stripped arm at 3 seeds plus the missing THIRD control seed. Registered: |stripped - control| <= s*_96 = 0.119, frozen from the three two-seed N=96 rungs that exist today. s* = 0.1484 is an N=48 number and is NOT reused there.

THE FALSIFIER THAT MATTERS, STATED BEFORE THE NUMBERS ARRIVE

If the stripped arm comes back HIGHER than the N=96 control by more than 0.119, the N=48 minimum does not transfer and every 'smallest architecture' claim in this programme gets relabelled as an N=48 claim -- which is a real possible outcome, not a formality. If any of the three stripped seeds CENSORS, the breaking point is the result and it becomes the headline. And finding 44's 1.64 plateau is an N=48 number: N=96's own plateau has never been measured, so no escape count will be quoted for F.

WHAT IS NOT BEING CLAIMED, AND THE OPEN N=48 QUESTION J LEFT

J4's sufficiency result is CORRELATIONAL and is a statement about this grid's range. 'One-dimensional' means no cut opened a second, deduction-specific way to fail; it does not mean these are the same model. The open N=48 question is J1's: the width curve is convex, so a fourth width point is the only way to know the shape of the d192-d256 stretch. That is deliberately NOT bundled into F, which changes the problem size and must change nothing else.

Stage J: 6 cells, 3 seeds per rung, 51280737, all COMPLETED 2026-11-06. Sufficiency refit over 9 scan dirs: 22/22 rungs, 47 cells, worst discriminating miss 0.0222 against bar 0.0423. Commit (uncommitted).